Reduction of methylene blue by glucose

Purpose

To observe the repeated oxidation and reduction of methylene blue and explain the chemical reactions of glucose and oxygen.

Materials

Glucose, methylene blue solution, sodium hydroxide, distilled water.

Procedure

- Combine glucose, sodium hydroxide solution, and water
- Add Methylene blue solution
- Cap the bottle and wait for the liquid to return to a colorless state.
- Shake the bottle and observe the change in color.

Observations

Beginning	
Middle	

End It starts to fade out gradually (10 min)	

Chemical explanation

In an alkaline solution, glucose transfers electrons to methylene blue(C16H18N3S +H2O +2e – C16H19N3S+OH), reducing it to a colorless state. Shaking dissolves oxygen from the air, which reoxidizes it and restores the blue color (C6H12O6 +3OH– C6H11O7 +2H_2O + 2e).Methylene blue acts as the redox mediator, transferring electrons from glucose to oxygen.